Informatica® Intelligent Data Management Cloud

October 2026

Monitoring Data Integration Jobs

Informatica Intelligent Data Management Cloud Monitoring Data Integration Jobs
October 2026

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Publication Date: 2026-10-16

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Preface

Use *Monitoring Data Integration Jobs* to learn how to view and monitor jobs that you start in Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Monitoring jobs

Monitor the jobs that you started in Data Integration on the **My Jobs** page. A job is an instance of an asset such as a mapping, task, or taskflow. Each time that you start the mapping, task, or taskflow, Data Integration creates a job to run it.

In Data Integration, you can monitor your jobs on the the following pages:

My Jobs page

Lists all jobs that were started by the currently logged in user. For most job types, you can stop and restart the job on this page. You can also download log files for some job types.

To open the **My Jobs** page, select **My Jobs**.

Job Details page

Displays detailed information about a specific job. You can view details for jobs that have completed or failed. For most job types, you can restart the job from this page. You can also download log files for some job types.

To view the details for a job, click the job name on the **My Jobs** page.

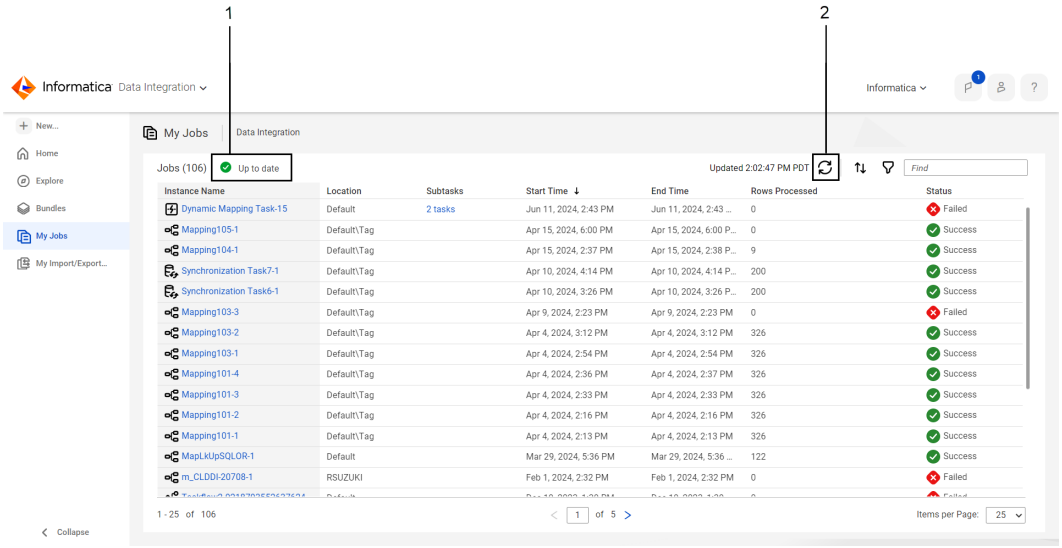
If you have the appropriate user role, you can also monitor your jobs on the **All Jobs** and **Running Jobs** pages in Monitor. For information about monitoring jobs on these pages, see the Monitor help.

Monitoring your jobs

You can monitor the jobs that you started on the **My Jobs** page. The **My Jobs** page lists your jobs that are currently running and your jobs that have completed.

The **My Jobs** page lists all the jobs that you ran within the last three days. If you have run less than 1000 jobs in the last three days, the log also includes older jobs, up to 1000 total jobs.

The following image shows the **My Jobs** page:



1. Status message that indicates whether information on the page is up-to-date or needs to be refreshed.
2. Refresh icon

To ensure that the information on this page is current, Data Integration polls the IDMC repository every five seconds. Information can become out-of-date when a job status changes or when a user starts a job.

The status message at the top of the page indicates whether the information on the page is up-to-date. If the information is out-of-date, the status message displays "Updates Available." To refresh the page, click the "Updates Available" message or the **Refresh** icon.

When a job completes, you can drill down on the job to view the job details. To drill down on a job, click the instance name.

Monitoring subtasks

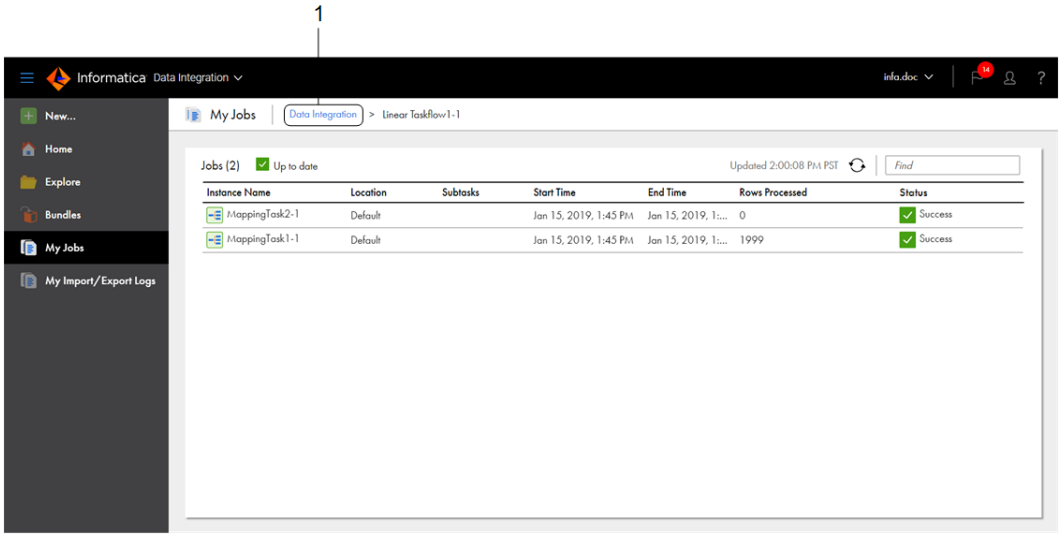
When you monitor a task instance that contains subtasks, you can also monitor the subtasks. When you monitor a taskflow instance that contains subtasks or subtaskflows, you can also monitor the subtasks and subtaskflows. You can monitor a subtask or subtaskflow from the **My Jobs** page in the service where you started the job.

To monitor subtasks or subtaskflows, open the **My Jobs** page, and then click the number of subtasks in the Subtasks column. For non-linear taskflows, you can also click **View Subtasks** in the Rows Processed column.

The following types of assets have subtasks:

- Mappings in advanced mode
- Replication tasks
- Taskflows and linear taskflows
- Data Profiling tasks
- Dynamic mapping tasks
- File listener
- File ingestion task

The following image shows the **My Jobs** page in Data Integration when you view subtasks for a linear taskflow:



1. Click the service name to close the subtasks view.

To close the subtasks view and return to the **My Jobs** page, click the service name in the menu at the top of the page. For example, to close the subtasks view for a linear taskflow, click **Data Integration** in the menu at the top of the page to the left of the parent task or taskflow instance name.

Job properties

The **My Jobs** page displays the job properties such as the name, start time, and status. You can right-click the column heading area to display or hide specific properties.

Each IDMC job is named `<asset name>-<instance number>`. For example, the first time you run mapping m_LoadCustOrders, the job is named m_LoadCustOrders-1. The second time you run the mapping, the job is named m_LoadCustOrders-2.

By default, the following properties are displayed for each job:

Property	Description
Instance Name	Name of the job in the following format: <asset name>-<instance number> For completed and failed jobs, you can click the instance name to view detailed information about the job.
Location	Project and folder path where the asset exists.
Subtasks	Displays the number of subtasks, when applicable. To view job properties for the subtasks, click the entry in this column.
Start Time	Date and time that the job was started.

Property	Description
End Time	Date and time that the job completed or stopped. Does not apply to running jobs.
Rows Processed	Total number of rows that the job has currently processed. This value includes the number of rows successfully written to the target as well as the number of error rows for each source, target, and transformation in the task. For jobs based on mappings in advanced mode, this total only includes rows processed by Snowflake V2, Redshift V2, and JDBC V2 connectors. If the job you are viewing is a non-linear taskflow instance, this field displays **View Subtasks**. Click **View Subtasks** to monitor the subtasks and subtaskflows for the taskflow. If the taskflow contains nested subtasks and subtaskflows, this field displays **View Subtasks** to view the nested subtasks and subtaskflows. Does not apply to advanced cluster subtasks.
Status	Job status. A job can have one of the following statuses: - Queued. The job is queued on a Secure Agent, but it has not started yet. Applies to subtasks of replication task and taskflow instances. - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by the user. Does not apply to data transfer tasks or dynamic mapping tasks. For subtasks of replication task instances, the parent job has stopped running, so the subtask cannot start. - Stopping. The job is stopping. Applies to tasks that are stopped cleanly. - Suspended. The job is paused. Applies to taskflow instances. - Warning. The job completed with errors. Applies to mapping and task instances. - Aborted. The job was aborted. Applies to file ingestion and replication task instances. - Failed. The job did not complete because it encountered errors. **Tip:** If the job status is Failed, you can hover the cursor over the job status to view and copy the error message.

You can also display the following additional properties by right-clicking in the column heading area:

Property	Description
Asset Name	Name of the asset that is associated with the job. For example, if the job is a mapping instance, this column displays the name of the associated mapping.
Asset Type	Type of asset that is associated with the job.
Runtime Environment	Runtime environment in which the job ran.
Duration	Amount of time the job ran before it completed or was stopped.
Success Rows	Number of rows successfully written to the target. Does not apply to file ingestion and replication tasks or advanced cluster subtasks.
Failure Rows	Number of rows that were not written to the target. Does not apply to file ingestion and replication tasks or advanced cluster subtasks.
Started By	Name of the user or schedule that started the job.
Throughput (Bytes/ Sec)	Number of bytes processed per second.

Property	Description
Bytes	Total number of bytes processed based on column metadata and number of rows.
Throughput (Rows/ Sec)	Number of rows processed per second from source to target, not including queue time.
Error Message	Error message, if any, that is associated with the job.

Customizing the jobs pages

You can decide which properties to display on the **My Jobs** page. You can also sort and filter jobs, drill down on a job to view detailed information about the job, and view properties for subtasks.

The following image shows the properties that are displayed by default on the **My Jobs** page:

1. Column heading area. Right-click this area to add and remove columns.
2. Sort icon
3. Filter icon
4. Find field

You can customize the jobs pages in the following ways:

Display, hide, or rearrange job properties.

To display or hide specific properties, right-click the column heading area and check or uncheck the properties.

To rearrange the columns, click a column heading and drag it to a different location.

Sort jobs.

To sort the displayed jobs, click the column heading for the property that you want to sort by. For example, to list the most recently completed jobs first, click the End Time column. The arrow in the column heading indicates the sort order, either ascending or descending. To reverse the sort order, click the column heading again.

You can also sort jobs by clicking the **Sort** icon and selecting the column name.

Find jobs.

To find specific jobs on the **My Jobs** page, use the find field. You can search for jobs by entering a full or partial job name or error message in the find field.

Filter jobs.

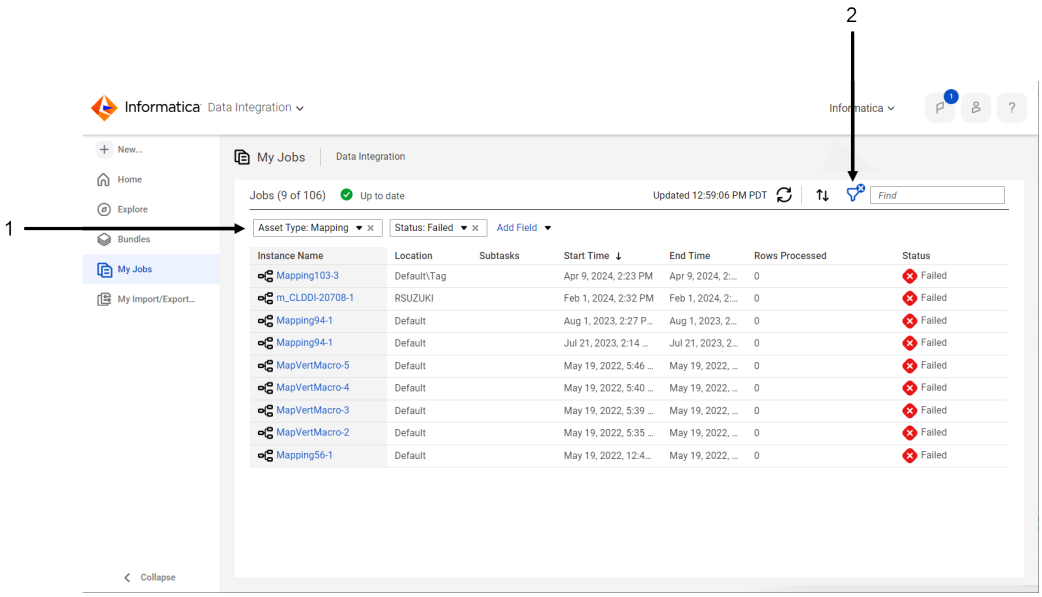
To filter the jobs that appear on the **My Jobs** page, click the **Filter** icon. Use filters to find specific jobs. You can specify keywords and partial strings in your filters.

To specify a filter, click **Add Field**, select the property to filter by, and then enter the property value. For example, to find jobs with names that contain the string "Sales," select "Asset Name" as the filter field, and enter "Sales" as the value. Or, to find jobs with errors that contain the string "Hosted Agent," select "Error Message" as the filter field, and enter "Hosted Agent" as the value.

You can specify multiple filters. For example, to find Data Integration failed mappings, select the following filter fields and values:

- Asset Type: Mapping
- Status: Failed

The following image shows the **My Jobs** page with the filters applied:



1. Applied filters
2. Remove Filter icon

To remove all applied filters, click the **Remove Filter** icon.

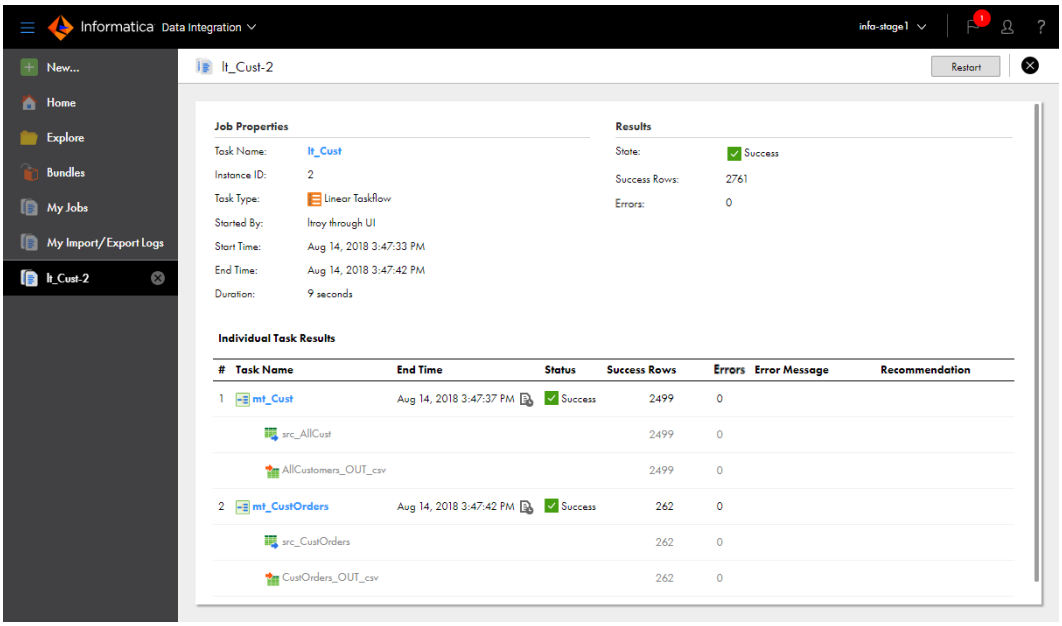
Viewing details for a specific job

You can drill down on any job on the **My Jobs** page to display detailed information about the job.

To drill down and view details about a specific job, click the job name.

To view details for jobs with subtasks, open the subtasks view, and click the subtask name.

The following image shows the details for a linear taskflow instance after you drill down:



The details displayed for the job vary based on the job type.

CHAPTER 2

Monitoring Data Integration jobs

You can monitor the Data Integration jobs that are running or have run. A job is an instance of a mapping, task, or taskflow. Each time that you start a mapping, task, or taskflow, Data Integration creates a job to run it.

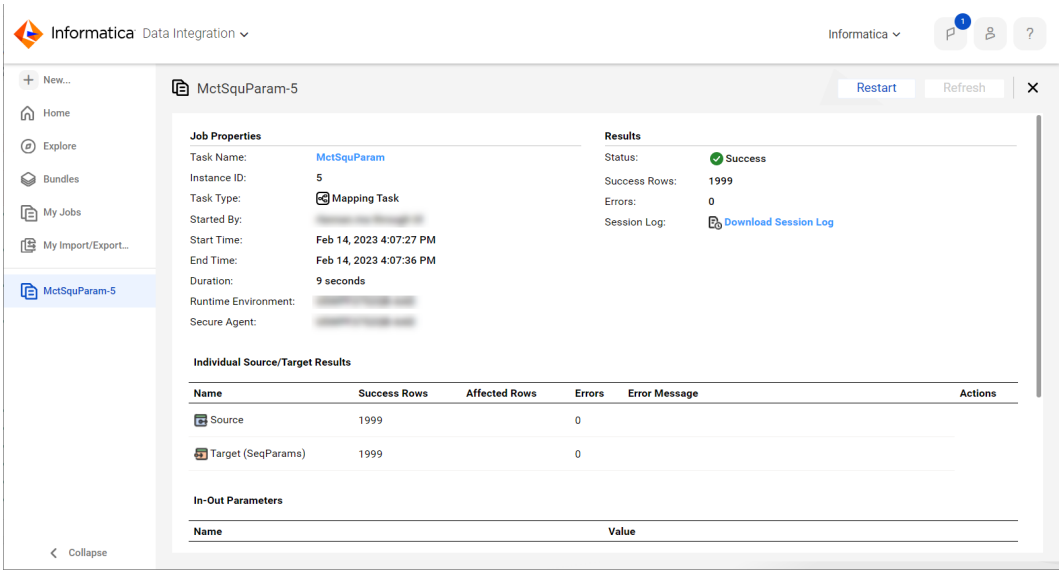
You can monitor the following types of Data Integration jobs:

- Mappings
- Mapping data preview jobs
- Mapping tasks
- Mappings in advanced mode
- Subtasks that run on the Data Integration Server
- Subtasks that run on an advanced cluster
- Dynamic mapping tasks
- Synchronization tasks
- Data transfer tasks
- Masking tasks
- Replication tasks
- File ingestion and replication tasks
- PowerCenter tasks
- Linear taskflows
- Advanced taskflows

Monitoring mappings and mapping tasks

To view detailed information about a specific mapping instance, task instance, or mapping data preview job, click the job name on the **My Jobs** page.

The following image shows the job details for a mapping task:



You view different details in each panel, and the details change based on the type of task that you run.

Job properties

The job properties for each mapping instance, task instance, or mapping data preview job display general properties about the instance.

The job properties include the following properties:

Property	Description
Task name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task type	Task type, for example mapping task.
Started by	Name of the user or schedule that started the job.
Start time	Date and time that the job was started.
End time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime environment	Runtime environment in which the job ran.
Secure Agent	Secure Agent that ran the job.

Job results

The job results for each mapping instance, task instance, or mapping data preview job display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you want to retain session logs for more than 10 runs, you can configure the **Maximum Number of Log Files** property in the task wizard. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Individual source and target results

Individual source and target results are displayed for mapping task instances and mapping data preview jobs that do not apply to mappings in advanced mode. Individual source and target results display details for individual sources and targets.

The individual source and target details include the following properties for each source and target:

Property	Description
Name	Name of the source transformation or name of the target transformation and the target object.
Success rows	Number of rows successfully written to the target.
Affected rows	Number of rows affected by the defined task operation. Doesn't apply to jobs based on mappings configured for full SQL ELT optimization or mappings in SQL ELT mode.

Property	Description
Errors	Total number of source error rows, target error rows, and transformation errors.
Error message	Error message, if any, that is associated with the source or target. Doesn't apply to jobs based on mappings configured for full SQL ELT optimization or mappings in SQL ELT mode.
Actions	Actions that you can take regarding the job.

Parameter details

Parameter details are displayed for job instances that contain in-out parameters or connection or object parameters that were overridden at runtime.

If the job instance includes in-out parameters, the job details display the value for each parameter after the task has completed.

For mapping task instances that include connection or object parameters that were overridden at runtime, the job details also display the location of the parameter file and the value of each parameter after the task has completed.

Sequence generator details

Sequence generator details are shown for mapping task and mapping data preview job instances that contain a Sequence Generator transformation that does not use a shared sequence.

The following table describes the properties shown for each sequence generator:

Property	Description
Transformation Name	Name of the Sequence Generator transformation.
Next Value	Value of the NEXTVAL output field in the Sequence Generator transformation.

SQL ELT optimization details

SQL ELT optimization details are shown for mapping task instances where the task is configured for SQL ELT optimization.

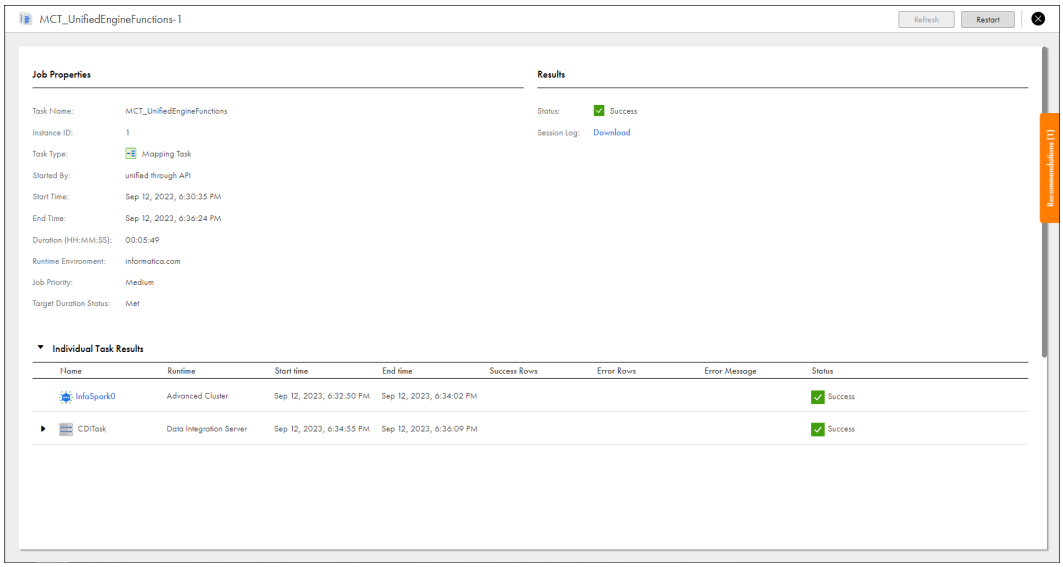
Expand the **SQL ELT Optimization** area to view SQL ELT details.

The **SQL ELT Optimization** area displays the SQL ELT type configured in the task and the type of SQL ELT the task executed. The area also displays the SQL queries pushed to the source or target, along with any errors.

Monitoring mappings and tasks in advanced mode

To view detailed information about a mapping in advanced mode, such as a mapping instance, task instance, or data preview job, click the job name on the **My Jobs** page.

The following image shows the job details for a mapping in advanced mode:



Job properties

The job properties display general properties about the instance for each mapping instance, task instance, or data preview job in advanced mode.

The following table describes the job properties:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, such as Mapping (Advanced Mode).
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.

Property	Description
Job Priority	Priority used to schedule the job.
Target Duration Status	Indicates whether the job met or exceeded its target duration.

Job results

The job results display the status of the job, a download link to the session log, and an error message for each mapping instance, task instance, or data preview job in advanced mode.

The following table describes the job results:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Success. The job completed successfully. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you want to retain session logs for more than 10 runs, you can configure the **Maximum Number of Log Files** property in the task wizard. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Individual task results

The individual task results display results for subtasks that process the mapping in advanced mode.

Each subtask runs either on the Data Integration Server or on an advanced cluster:

- **Data Integration Server.** You can expand the Data Integration Server subtask to view the source and target results.
- **Advanced cluster.** You can drill down on the advanced cluster subtask. For more information, see "Monitoring advanced cluster subtasks" on page 24.

You can hover over each row in the individual task results to download a detailed log for Data Integration Server subtasks or to download the agent and driver logs for advanced cluster subtasks.

The following table describes the individual task results:

Property	Description
Name	Name of the subtask.
Runtime	Indicates whether the subtask ran on the Data Integration Server or on an advanced cluster.
Start time	Date and time that the subtask started.
End time	Date and time that the subtask completed or stopped.
Success Rows	Number of rows successfully written to the target. For each source and target in the subtask, this field displays the number of rows successfully read from the source and the number of rows successfully written to the target. For advanced cluster subtasks, this field displays a value only if the task runs using SQL ELT optimization. For Spark subtasks, this field displays a value only for supported connectors.
Error Rows	Total number of source error rows, target error rows, and transformation errors. For each source and target in the subtask, this field displays the number of rows that were not read from the source and the number of rows that were not written to the target. For advanced cluster subtasks, this field displays a value only if the task runs using SQL ELT optimization. For Spark subtasks, this field displays a value only for supported connectors.
Error Message	Error message, if any, that is associated with the subtask.
Status	Status of the subtask. A subtask can have one of the following statuses: - Queued. The subtask is queued on a Secure Agent, but it has not started yet. - Starting. The subtask is starting. - Running. The subtask is still running. - Success. The subtask completed successfully. - Stopped. The parent task stopped running, so the subtask cannot start. - Warning. The subtask completed with errors. - Failed. The subtask did not complete because it encountered errors.

Runtime plan

The runtime plan is a visualization that closely approximates how the data actually flows through the mapping at runtime. You can view which mapping logic runs on the Data Integration Server or on an advanced cluster.

Note: If Monitor fails to retrieve the runtime plan, download the session log and review the mapping compilation log in the archive file. You can search for the following line: `[LDTM_7343] Generating runtime plan`.

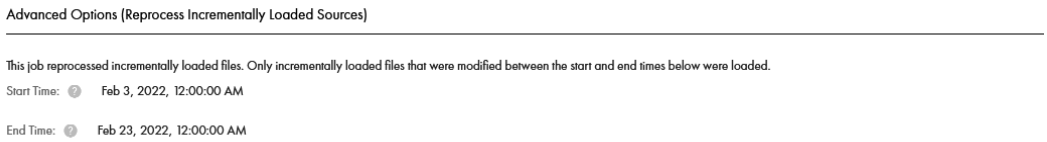
For more information about runtime plans, see *Mappings*.

Reprocessing job details

If you run a job to reprocess incrementally-loaded source files, you can view details about the reprocessing job.

You can view the start and end times of the time range that the job uses to search for new and modified files in the configured source directories. The start time is the time that you configure in the advanced options. If you configure reprocessing in a time interval, the end time is time that you configure in the advanced options. Otherwise, the time at which the job runs determines the end time.

The following image shows an example of the monitoring details for a reprocessing job:



Note: If you restart a job that reprocessed incrementally-loaded source files, the new job runs as a regular job without the advanced options.

Incremental file load details

If you configure sources to incrementally load files, you can view details about the file load.

You can view the start and end times of the time range that the job uses to search for new and modified files in a directory.

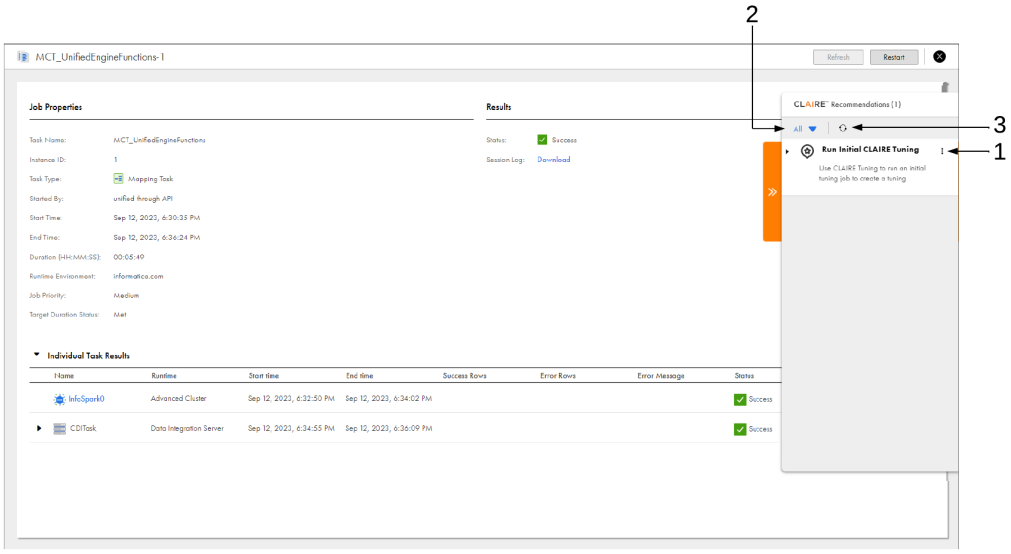
The start load time is derived from the end load time of the previous job. The start load time does not apply when the job runs for the first time or you reset the last load time in the mapping task.

Jobs process all files modified up to the end load time, including up to 999 milliseconds after the second. For example, if the end load time of a job appears as Aug 28, 2019 2:15:22 p.m., the job processes all files modified up to Aug 28, 2019 2:15:22.999 p.m.

CLAIRE recommendations

View CLAIRE recommendations to improve job performance and to reduce job costs in the **Recommendations** panel. CLAIRE recommendations are available if CLAIRE is enabled in your organization.

The following image shows the **Recommendations** panel:



1. Actions menu

Use the **Actions** menu to mark recommendations as complete or incomplete, or to opt in and opt out of recommendations.

CLAIRE automatically applies some recommendations. You can use the **Actions** menu to opt out of the recommendation or opt back in.

Other recommendations require manual action. These recommendations appear as to-do items. You can use the **Actions** menu to mark to-do items as complete, or mark them as incomplete. You can also opt out of the recommendation or opt back in.

2. Filter recommendations

Use the **Filter** menu to filter recommendations. You can use the following filters:

- **All** shows all recommendations.
- **To Do** shows recommendations that require manual action.
- **Applied** shows recommendations that are applied automatically and recommendations that are marked complete.
- **Opted-Out** shows recommendations that are opted out of.

3. Refresh recommendations

Refresh the recommendations to update the recommendations in the **Recommendations** panel.

Monitoring advanced cluster subtasks

To view detailed information about a subtask that runs on an advanced cluster, navigate to the subtask from the individual task results for a mapping in advanced mode.

Note: When you monitor a subtask that runs on an advanced cluster while the job is running, you must refresh the page to view updates to the job properties, job results, and Spark task details.

Job properties

The job properties for each subtask that runs on an advanced cluster display general properties about the instance.

The following table describes the job properties for subtasks that run on an advanced cluster:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this case, mapping task.
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Advanced configuration	Advanced configuration that was used to create the advanced cluster.
Cluster	Advanced cluster where the job runs. You can click the cluster name to navigate directly to the monitoring details for the cluster.

Job results

The job results for each subtask that runs on an advanced cluster display the status of the job, a download link to the Spark execution plan, and an error message, if any.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is either queued or running. - Success. The job completed successfully. - Failed. The job did not complete because it encountered errors. If the advanced cluster is not running when you run a job, the job waits for the cluster to start. During this time the job status is Starting. If the Secure Agent fails while the job is running, the status of the job continues to display Running. You must cancel the job and run the job again. The status for a queued job displays Running. To find out if a job is queued or running, check the session log.
Execution Plan	Allows you to download the Spark execution plan which shows the runtime Scala code that the advanced cluster uses to run the data logic in the mapping. You can use the Scala code to debug issues in the mapping.
Error Message	Error message, if any, that is associated with the job.

Job results for tuning

If you tune a mapping task that runs on an advanced cluster, the job results show the tuning job status and a link to the subtasks.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Running. The job is either queued or running. - Success. The job completed successfully. - Stopped. The job was stopped. - Failed. The job did not complete because it encountered errors. If the Secure Agent fails while one of the subtasks is running, the statuses of the subtask and the tuning job display Running. You must stop the tuning job and configure tuning from the mapping task details again. The status for a queued job displays Running. To find out if a job is queued or running, check the session log.
Subtasks	Number of subtasks that are part of the tuning job. Each subtask represents a run of the mapping task. If a link is available, click the link to monitor each mapping task.
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.

Property	Description
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Spark task details

In subtasks that run on an advanced cluster, the mapping is translated into Spark tasks that process the data logic simultaneously. You can view details for each Spark task listed under **<Spark task name> Task Results**.

The following table describes the details for each Spark task:

Property	Description
Status	Status of the Spark task. The Spark task can have one of the following statuses: - Running. The task is running. - Succeeded. The task completed successfully. - Failed. The task did not complete because it encountered errors. - Stopped. The task was stopped. - Unknown. The status of the task is unknown. If the Secure Agent fails while the job is running, the status of the Spark tasks continues to display Running. You must cancel the job and run the job again.
Start time	Date and time when the Spark task started.
End time	Date and time when the Spark task ended.
Duration	Amount of time that the Spark task ran.
Memory Per Executor	Amount of memory that each Spark executor uses.
Cores Per Executor	Number of cores that each Spark executor uses.
Driver and Agent Job Logs	Select **Download** to download the Spark driver and agent job logs.
Advanced Log Location	The log location that is configured in the advanced configuration for the advanced cluster. You can navigate to the advanced log location to view and download the agent job log, Spark driver log, and Spark executor logs.

Each Spark task is translated into Spark jobs, which are further broken down into stages. You can view the following details for each Spark job and stage:

Property	Description
Job Name	Name of the Spark job or stage.
Start time	Date and time when the Spark job or stage started. Start time might be "NA" for aborted tasks.

Property	Description
End time	Date and time when the Spark job or stage ended. End time might be "NA" for aborted tasks.
Duration	Amount of time that the Spark job or stage ran.
Total Tasks	Number of tasks the Spark job or stage attempted.
Successful Tasks	Number of tasks the Spark job or stage successfully completed.
Failed Tasks	Number of tasks that the Spark job or stage failed to complete.
Running Tasks	Number of tasks that the Spark job or stage is currently running.
Input Size / Records	Size of the file and number of records input by the Spark job or stage.
Output Size / Records	Size of the file and number of records output by the Spark job or stage.
Status	Status of the Spark job or stage. The status can be one of the following values: - Running. The job or stage is running. - Success. The job or stage completed successfully. - Failed. The job or stage did not complete because it encountered errors. - Aborted. The job or stage did not complete because the user aborted the mapping task. **Note:** After you abort a mapping task, there might be some lag time before the Monitor service shows the status as *Aborted*.

Monitoring code tasks

To view detailed information about a specific code task, click the instance name on the **My Jobs** page.

Job properties

The job properties for each code task instance display general properties about the task.

The following table describes the job properties for the code task:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this case, code task.
CodeTask ID	Code task unique identifier.
Started By	Name of the user that started the job.
Start Time	Date and time that the job was started.

Property	Description
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Advanced configuration	Advanced configuration that was used to create the advanced cluster.
Cluster	Advanced cluster where the advanced job runs. You can click the cluster name to navigate directly to the monitoring details for the cluster.

Job results

The job results for each code task instance display the status of the job and success and error statistics.

The following table describes the job results for the code task:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is either queued or running. - Success. The job completed successfully. - Failed. The job did not complete because it encountered errors.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Code task API execution parameters

The execution parameters for each code task instance display the API parameters used in the task.

The following table describes the execution parameters for the code task:

Property	Required / Optional	Description
Override Code Task Timeout	Optional	Overrides the code task timeout value for this execution. A value of -1 signifies no timeout.
Log Level	Optional	Log level for session logs, agent job log, Spark driver, and executor logs. Valid values are: none, terse, normal, verboseInitialization, or verboseData. The default value is normal.

The following table describes the Spark properties for the code task:

Property	Required / Optional	Description
Main Class	Required	Entry point of the Spark application. For example: org.apache.spark.examples.company.SparkExampleApp
Main Class Arguments	Optional	Ordered arguments sent to the Spark application main class. For example: --appTypeSPARK_PI_FILES_JARS-- classesToLoadcom.company.test.SparkTest1Class
Primary Resource	Required	Scala JAR file that contains the code task.
JAR File Path	Optional	The directory and file name of the JAR file that is uploaded to the cluster and added to the Spark driver and executor classpaths.
Spark File Path	Optional	The directory and file name of the Spark file that is uploaded to the cluster and available under the current working directory.
Custom Properties	Optional	Spark properties or other custom properties that Data Integration uses.

Spark application task details

The Spark application task details for each code task display under **Spark Application Task Results**.

Each Spark application task includes the following details:

Property	Description
Status	Status of the Spark task. The Spark task can have one of the following statuses: - Running. The task is running. - Success. The task completed successfully. - Failed. The task did not complete because it encountered errors. - Stopped. The task was stopped. - Unknown. The status of the task is unknown. If the Secure Agent fails while the job is running, the status of the Spark tasks continues to display Running. You must cancel the job and run the job again.
Start time	Date and time when the Spark task started.
End time	Date and time when the Spark task ended.
Duration	Amount of time that the Spark task ran.
Memory Per Executor	Amount of memory that each Spark executor uses.
Cores Per Executor	Number of cores that each Spark executor uses.
Driver and Agent Job Logs	Select **Download** to download the Spark driver and agent job logs.
Advanced Log Location	The log location that is configured in the advanced configuration for the advanced cluster. You can navigate to the advanced log location to view and download the agent job log, Spark driver log, and Spark executor logs.
Error Message	Error message, if any, that is associated with the job.

Each Spark application task is translated into Spark jobs, which are further broken down into stages. You can view the following details for each Spark job and stage:

Property	Description
Job Name	Name of the Spark job or stage.
Duration	Amount of time that the Spark job or stage ran.
Total Tasks	Number of tasks the Spark job or stage attempted.
Failed Tasks	Number of tasks that the Spark job or stage failed to complete.
Input Size / Records	Size of the file and number of records input by the Spark job or stage.

Property	Description
Output Size / Records	Size of the file and number of records output by the Spark job or stage.
Status	Status of the Spark job or stage. The status can be one of the following values: - Running. The job or stage is running. - Success. The job or stage completed successfully. - Failed. The job or stage did not complete because it encountered errors. - Aborted. The job or stage did not complete because the user aborted the code task. **Note:** After you abort a code task, there might be some lag time before the Monitor service shows the status as *Aborted*.

Monitoring dynamic mapping tasks

To view detailed information about a specific dynamic mapping task, click the instance name on the **My Jobs** page.

When you view details for a completed or running task, you can restart the task run from the beginning. If one or more jobs in the task failed, you can resume the task run. When you resume a dynamic mapping task, Data Integration reruns jobs that failed or were skipped during the previous run. If the task run was stopped due to a warning and the task is configured to stop on warning, Data Integration also reruns jobs that completed with warnings.

You can also restart or resume a dynamic mapping task.

Job properties

The job properties for each dynamic mapping task include general properties about the task.

The job properties include the following properties:

Property	Description
Task name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the fourth run of a task, this field displays "4."
Task type	Task type, for example, dynamic mapping task.
Started by	Name of the user or schedule that started the job.
Start time	Date and time that the job was started.
End time	Date and time that the job completed or stopped.
Duration	Amount of time that the job ran before it completed or stopped.
Runtime Environment	Runtime environment in which the job ran.

Job results

The job results for each dynamic mapping task display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target..
Error Rows	Total number of source error rows, target error rows, and transformation errors.

Individual job results

The individual job results display results for jobs in the dynamic mapping task instance, and details for individual sources and targets.

For each job, the job details include the following properties:

Property	Description
Name	Name of the job, source transformation, or target object.
Group	Group the job is assigned to in the dynamic mapping task.
Start Time	Date and time that the job started.
End Time	Date and time that the job completed or stopped.
Success Rows	Number of rows successfully written to the target.
Affected Rows	Number of rows affected by the defined task operation.
Error Rows	Number of rows not written to the target.
Error Message	Error message, if any, that is associated with the job.
Status	Job status.

You can restart an individual job in a dynamic mapping task from the **My jobs** page. To restart a specific job, in the row that contains the job, click **Restart**.

To see detailed information about each job, click the job name. Use **Refresh** to get the most current information and status for each job.

Monitoring synchronization tasks

To view detailed information about a specific synchronization task instance, click the job name on the **My Jobs** page.

Job properties

The job properties for the synchronization task display general properties about the instance.

The job properties include the following properties:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this instance, synchronization task.
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Secure Agent	Secure Agent that ran the job.

Job results

The job results for each synchronization task display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Success. The job completed successfully. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.

Property	Description
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you want to retain session logs for more than 10 runs, you can configure the **Maximum Number of Log Files** property in the task wizard. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Error Rows File	Allows you to preview the error rows file for instances that have error rows. Click **Preview Error Rows File** to download the first 25 error rows and first 50 columns as a CSV file.
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Monitoring data transfer tasks

To view detailed information about a specific data transfer task, click the instance name on the **My Jobs** page.

Job properties

The job properties for each data transfer task include general properties about the task.

The job properties include the following properties:

Properties	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, for example, data transfer task.
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.

Properties	Description
Secure Agent	Secure Agent that ran the job.
Context Name	Name of the task that is associated with the job.

Job results

The job results for each data transfer task display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you want to retain session logs for more than 10 runs, you can configure the **Maximum Number of Log Files** property in the task wizard.

Individual source and target results

Individual source and target results display details for individual sources and targets.

The individual source and target details include the following properties for each source and target:

Property	Description
Name	Name of the source or target object.
Success rows	Number of rows successfully written to the target.
Affected rows	Number of rows affected by the defined task operation.
Errors	Total number of source error rows or target error rows.
Error Message	Error message, if any, that is associated with the job.

Monitoring masking tasks

To view detailed information about a specific masking task instance, click the job name on the **My Jobs** page.

Job properties

The job properties for each masking task instance display general properties about the instance.

The job properties include the following properties:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this instance, masking task.
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Secure Agent	Secure Agent that ran the job.

Job results

The job results for each masking task instance display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.

Property	Description
Error Rows File	Allows you to preview the error rows file. Click **Preview Error Rows File** to download the first 25 error rows and first 50 columns as a CSV file.
Subset Rows	Number data subset rows loaded to the target.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you need the session logs for earlier runs, take a backup of the directory that holds the session log files. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Activity log details

The activity log displays results for individual subtasks.

The activity log details include the following properties for each subtask:

Property	Description
Name	Name of the subtask
End time	Date and time the subtask completed. You can also download the session log for the subtask.
Status	Status of the subtask.
Success rows	Number of rows successfully written to the target.
Error rows	Number of rows that were not written to the target
Error message	Error message, if any, associated with the job.
Recommendation	Recommended action you can take regarding the job.

Monitoring replication tasks

To view detailed information about a specific replication task instance, click the job name on the **My Jobs** page.

Job properties

The job properties for each replication task display general properties about the instance.

The job properties include the following properties:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this instance, replication task.
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Secure Agent	Secure Agent that ran the job.

Job results

The job results for each replication task display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Queued. The job is queued on a Secure Agent, but it has not started yet. - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.

Property	Description
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Individual object results

The individual object results display results for individual target objects. You can also download the session logs for individual subtasks.

The job details include the following properties for each target object:

Property	Description
Object name	Name of the target object.
End time	Date and time that the job completed or stopped.
Status	Job status, either running, success, warning, or failed.
Success rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.
Error message	Error encountered when writing to the target, if any.
Recommendation	Recommended action for errors.

Monitoring PowerCenter tasks

To view detailed information about a specific PowerCenter task, click the job name on the **My Jobs** page.

Job properties

The job properties for each PowerCenter task instance display general properties about the instance.

The job properties include the following properties:

Property	Description
Task Name	Name of the task.
Instance ID	Instance number for the task. For example, if you are looking at the third run of the task, this field displays "3."
Task Type	Task type, in this instance, PowerCenter.

Property	Description
Started By	Name of the user or schedule that started the job.
Start Time	Date and time that the job was started.
End Time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.
Runtime Environment	Runtime environment in which the job ran.
Secure Agent	Secure Agent that ran the job.

Job results

The job results for each PowerCenter task instance display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success Rows	Number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors.
Session Log	Allows you to download the session log file. By default, IDMC stores session logs for 10 runs before it overwrites the logs with the latest runs. If you need the session logs for earlier runs, take a backup of the directory that holds the session log files. Session log files are written to the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/logs`
Requested Compute Units Per Hour	Number of serverless compute units per hour that the task requested. You can view the number of requested compute units if the task runs in a serverless runtime environment.
Total Consumed Compute Units	Total number of serverless compute units that the task consumed. You can view the number of consumed compute units if the task runs in a serverless runtime environment.
Error Message	Error message, if any, that is associated with the job.

Monitoring taskflows

You can monitor the details of the taskflows that are running or have run in your organization on the **My Jobs** page. The **Started By** column displays the name of the user who first started the taskflow even if the taskflow is restarted by another user.

Note: To view taskflow run details on the Monitor page, you need Read permission on the taskflow. To stop a running taskflow, you need Execute permission on the taskflow.

The following table lists the actions available at each privilege level:

Action	Read Privilege	Execute Privilege
Abort	No	Yes
Download logs	Yes	Yes
Run or re-run	No	Yes
View execution listing	Yes	Yes
View run details	Yes	Yes
Schedule	No	Yes
Stop	No	Yes

The taskflow instance name appears in the following format:

`<taskflow name>-<runID>`

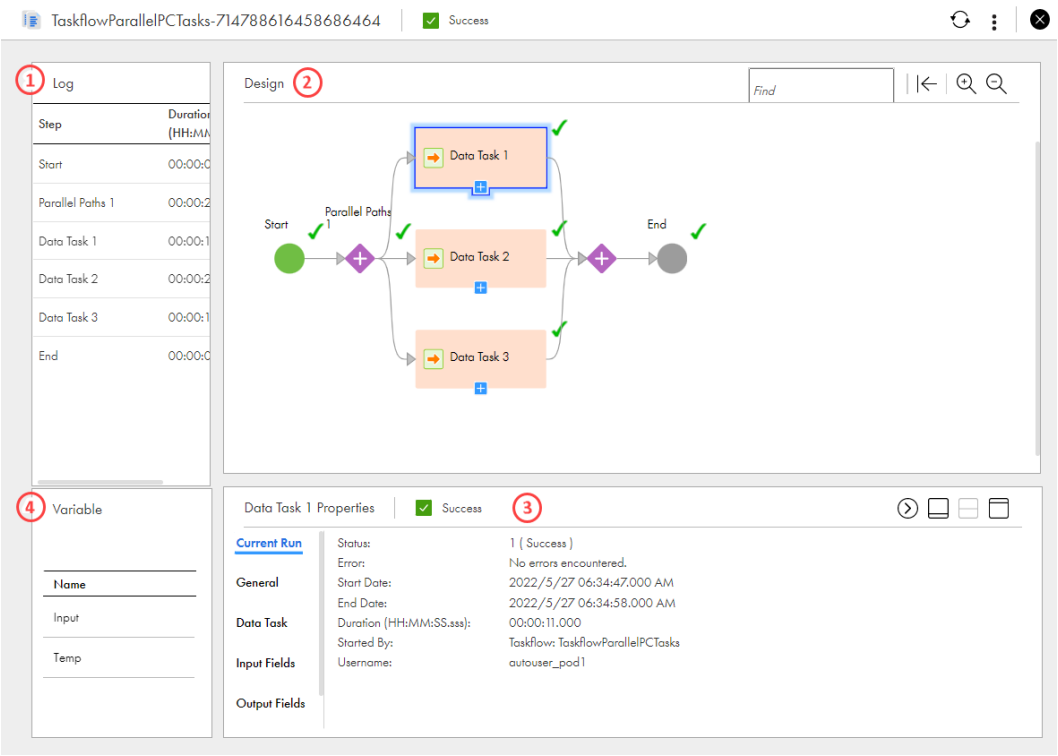
If you had added a custom name to a taskflow name by using an API or the RunAJob utility, the taskflow name appears in the following format:

`<taskflow name>-<custom name>-<runID>`

To view detailed information about a taskflow instance, click the job name on the **My Jobs** page.

Note: You can view taskflows with invalid query expressions on the **My Jobs** page. If no error appears on the start step of the taskflow instance page, refresh the page or download the log.

The following image shows the job details for a taskflow instance:



1. Log panel. A chronological list of the activities that the taskflow has run.
2. Design panel. A canvas view of the taskflow.
3. Properties panel. The properties of the taskflow element that you select on the canvas. Click the Start step to view properties specific to the current taskflow instance. Click a Data Task step to see properties specific to the task.
4. Variable. The input and temp variables used to run the taskflow instance. To update the variables and their associated values, click the **Edit** icon next to the Input or Temp variable in the Variable panel.

Start step properties

The Start step properties display information about the current run of a taskflow.

When you click the Start step, the properties panel displays the following information:

Property	Description
Current Run	You see the following properties: - Instance ID. A unique number that identifies the current taskflow instance. - Status. The state that the taskflow instance is in. Taskflows can be in the Running, Failed, Success or Suspended states. - Last Fault Name. The cause for the fault occurrence. This field appears only when the taskflow fails. - Error. Exception or error message. - Start Date. The date and time at which the taskflow instance was started. - End Date. The date and time at which the taskflow instance ended. You see an End Time value if a taskflow is in the Success or Failed states. - Duration. The amount of time the taskflow instance ran before it completed. You see a Duration value if a taskflow is in the Success or Failed states. - Username. The name of the user who ran this instance of the taskflow. - Runtime Environment. The runtime environment in which the taskflow ran. You can run taskflows only in the Cloud environment.
Input Fields	Displays the name and value provided as input for the current taskflow instance.

Data Task properties

The Data Task step properties display details about the current run of the data task of a taskflow.

When you click a data task, the properties panel displays the following information:

Property	Description
Current Run	You see the following properties: - Status. Indicates the status of the data integration task. A data integration task can be in the **0 (Faulted)**, **1 (Success)**, or **2 (Warning)** states. - Error. If the data integration task is in the **0 (Faulted)** or **2 (Warning)** states, the error message appears here. - Start Date. The date and time that the data integration task started. - End Date. The date and time that the data integration task ended or was suspended. - Duration. The amount of time the data integration task ran before it completed or was suspended. - Started By. The name of the taskflow that called the data integration task. - Username. The name of the user who ran this instance of the taskflow.
General	The name and type of the Data Task step.
Data Task	You see the following properties: - Name. The name of the data integration task. - Type. Specifies whether the data integration task is a data transfer task, dynamic mapping task, mapping task, PowerCenter task, or synchronization task.

Property	Description
Input Fields	Properties specific to the data integration task that you added to a Data Task step. You see the following properties: - Wait for Task to Complete. Indicates if the Data Task step waits for the data integration task to complete. This value is always **Yes**. - Max Wait (Seconds). The maximum length of time in seconds that the Data Task step waits for the data integration task to complete. - Task Name. The name of the data integration task. - GUID. The unique identifying number of the data integration task. - Task Type. Indicates whether the data integration task is a data transfer task, dynamic mapping task, mapping task, PowerCenter task, or synchronization task. - Has Inout Parameters. Indicates whether the data integration task contains input parameters or output parameters.
Output Fields	You see output field properties if the **State** of the data integration task is **1 (Success)**. Properties specific to the completed data integration task appear here. **Note:** If the data task is a mapping task that is based on a mapping in advanced mode, the output field properties include success target rows and error target rows, but they do not include success source rows and error source rows.
Fault	If the data integration task faulted, you see the following properties: - Code. Indicates whether the fault is an error or a warning. - Reason. Indicates the cause of the error or fault. - Detail. Provides details about the error or fault. You see the output field properties when the data integration task status is **3 (Failed)** and one of the following conditions are met: - The **On Error** field is set to **Ignore** or **Custom error handling**. - The **Fail taskflow on completion** option is set to **If this task fails**.

IntegrationOps Task properties

The IntegrationOps Task step properties display details of the current run of the IntegrationOps task in a taskflow. The taskflow continues to run the subsequent steps even if the IntegrationOps task fails.

When you click the IntegrationOps task, the **Properties** panel displays the following properties:

Property	Description
Input Fields	Properties specific to the process you added to the IntegrationOps Task step. Displays the following properties: - Task Name. The name of the Application Integration process. - GUID. The unique identifying number of the process. - Task Type. The task type. For an IntegrationOps Task, the task type is always PROCESS. - Agent Name. The Secure Agent name where the process ran.
Output Fields	Displays the output fields of the process along with their associated values. The output fields are displayed only if the IntegrationOps task completed successfully.
Fault	If the IntegrationOps task is faulted, you see the following properties: - Code. Indicates whether the fault is an error or a warning. - Reason. Indicates the cause of the error or fault. - Detail. Provides details about the error or fault.

Notification Task properties

The Notification Task step properties display details about the current run of the Notification Task of a taskflow.

When you click a Notification Task, the **Properties** panel displays the following information:

Property	Description
General	The name of the Notification Task step.
Details	Displays the following input fields of the Notification Task: - Email To. The primary recipients for the email notification. - Email Cc. The recipients who need to be sent a copy of the email notification. - Email Bcc. The additional recipients who need to be sent a copy of the email notification. The recipients in the **Email To** and **Email Cc** fields will not be able to see the recipients in the **Email Bcc** field. If the field contains more than one recipient, the recipients will not be able to see the other recipients in the **Email Bcc** field. - Email Subject. A short and descriptive subject that introduces the email. - Email Content Type. The type of formatting that the email content uses. - Email Body. The content that the email contains. If you selected the email body as **Content**, the email content appears within HTML tags. For example, `<p>Hello World</p>`

Command Task step properties

The Command Task step properties display details of the current run of the command task in a taskflow. When you click a command task in the taskflow, the **Properties** panel displays the following properties:

Property	Description
General	Displays the name of the command task.
Input Fields	Displays the value of the runtime environment, script file name, input arguments, and work directory that you configure for the command task. The fields also display the maximum length of time in seconds that the command task step waits for the task to complete.
Output Fields	Displays the value of the run ID, start and end time, exit code, execution status, and standard error. The output fields are displayed only if the command task executes successfully.
Fault	Displays the value of the run ID, start and end time, exit code, execution status, and standard error. The fault fields are displayed only if the command task fails due to a script failure. The details help you analyze the reason for the fault occurrence. You can then take an appropriate action on the faulted command task and proceed with the execution of the taskflow.

If the Command Task step contains multiple script files, you can view all the scripts in the taskflow instance as subtasks of the command task from the **My Jobs** page. If a script fails, the status of the script is set to failed. You can also download the log file to understand the reason for the script failure.

File Watch Task step properties

The File Watch Task step properties display details about the current run of the file watch task of a taskflow.

When you click a File Watch Task step, the **Properties** panel displays the following information:

Property	Description
Current Run	Displays the following details: - Status. Status of the file watch task. - Started By. Name of the taskflow that called the file watch task.
General	Displays the name of the File Watch Task step.
File Watch Task	Displays the name and type of the file listener that the File Watch Task step contains. The type is always set to **MI_FILE_LISTENER**.
Input Fields	Displays the following input fields: - Wait for Task to Complete. Indicates whether the File Watch Task step waits for the file listener to complete. This value is always set to **true**. - Max Wait. Maximum time in seconds that the File Watch Task step waits for the file listener to complete. This value is always set to **604800** seconds, which is 7 days. - Task Name. Name of the file listener. - GUID. Unique identification number for the file listener. - Task Type. Indicates the type of file listener that the File Watch Task step contains. This value is always set to **MI_FILE_LISTENER**. - serviceProviderId. Service provider ID for the file listener. This value is always set to **filelistener**. - serviceProviderContextPath. Service provider context path for the file listener. This value is always set to **filelistener/**.

Property	Description
Output Fields	Displays the following output fields if the File Watch Task step completed successfully: - id. ID for the File Watch Task step run. - monitorJobId. Job ID that you can use to check the status of the file listener in the **File Transfer Logs** tab of the Monitor service. Append the monitorJobId to the file listener name to find out the instance name in the **File Transfer Logs** tab of the Monitor service. For example, if the monitorJobId is **7500** and the name of the file listener is **FL_Arrive**, the instance name that you need to look for in the **File Transfer Logs** tab of the Monitor service would be **FL_Arrive-7500**. - notificationTime. Timestamp when the File Watch Task step completed. The timestamp includes the date and time up to seconds. - status. Status of the file watch task. This value is always set to **Completed**. - fileEvents. Provides a list of files along with the following details: - Event ID. Auto-generated ID for the event. - Event type. Indicates the nature of the file event. You see one of the following values: - ARRIVED - UPDATED - DELETED - File last modified. Timestamp when the file was last modified. The timestamp includes the date and time up to seconds. - File name. Name of the file that arrived, was updated, or deleted as part of the event. - File path. Name of the file and the path where the file exists. - File size. Size of the file in KB. - Task Id. This value is always blank. - Run Id. This field has the same value as the **monitorJobId** field.
Fault	If the file watch task faulted, you see the following properties: - Code. Indicates whether the fault is an error or a warning. - Reason. Cause of the error or fault. - Details. Details about the error or fault.

Consider the following rules and guidelines when you monitor the File Watch Task step:

- The **All Jobs** page does not display the file watch task as a subtask. It always displays the number of rows processed as **0** if the taskflow includes a File Watch Task step.
- If the file watch task faulted and the **On Error** property was set to **Suspend Taskflow**, no information is displayed in the **Properties** panel.

Ingestion Task step properties

The Ingestion Task step properties display details about the current run of the Ingestion task of a taskflow.

When you click an Ingestion Task step, the **Properties** panel displays the following information:

Property	Description
Current Run	Displays the following details: - Status. Status of the file ingestion task. - Start Date. Date and time when the file ingestion task started. - End Date. Date and time when the file ingestion task ended or was suspended. - Duration. Amount of time in seconds that the file ingestion task ran before it completed or was suspended. - Started By. Name of the taskflow that called the file ingestion task.
General	Displays the name of the Ingestion Task step.
Ingestion Task	Displays the name and type of the ingestion task that the Ingestion Task step contains. The type is set to **MI_TASK** for file ingestion tasks.
Input Fields	Displays the following input fields: - Wait for Task to Complete. Indicates whether the Ingestion Task step waits for the file ingestion task to complete. This value is always set to **true**. - Max Wait. Maximum time in seconds that the Ingestion Task step waits for the file ingestion task to complete. This value is always set to **604800** seconds, which is 7 days. - Task Name. Name of the file ingestion task. - GUID. Unique identification number for the file ingestion task. - Task Type. Indicates the type of task that the Ingestion Task step contains. This value is always set to **MI_TASK** for file ingestion tasks. - serviceProviderId. Service provider ID for the file ingestion task. This value is always set to **filelistener**. - serviceProviderContextPath. Service provider context path for the file ingestion task. This value is always set to **mitask/**.
Output Fields	Displays the following output fields if the Ingestion Task step completed successfully: - endTime. Date and time when the file ingestion task completed. - failedFiles. Number of failed files in the file ingestion task run. - jobNumber. ID for the file ingestion task run. - messageText. Indicates whether the job completed normally. This value is always set to **Job completed normally**. - startTime. Date and time when the file ingestion task started. - status. Indicates whether the Ingestion Task step completed. This value is always set to **Success**. - successFiles. Number of files that were successfully processed in the file ingestion task. - fileDetails. Provides a list of objects with the following details: - Duration. Amount of time in seconds that the file ingestion task ran before it completed. - Last modified. Date and time when the file was last modified. - Path. Name of the file and the path where the file exists. - Size. Size of the file in KB. - Status. Indicates the status of the file transfer. This value is always set to **Success**. - Transfer direction. Indicates the transfer direction for the file. You see one of the following values: - DOWNLOAD. The file was downloaded from the source. - UPLOAD. The file was uploaded to the target.
Fault	If the ingestion task faulted, you see the following properties: - Code. Indicates whether the fault is an error or a warning. - Reason. Cause of the error or fault.

Consider the following rules and guidelines when you monitor the Ingestion Task step:

- The **All Jobs** page does not display the ingestion task as a subtask. It always displays the number of rows processed as **0** if the taskflow includes an Ingestion Task step.
- If the ingestion task faulted and the **On Error** property was set to **Suspend Taskflow**, no information is displayed in the **Properties** panel.

Subtaskflow step properties

The Subtaskflow step properties display details about the current run of the subtaskflow.

When you click a Subtaskflow step, the **Properties** panel displays the following information:

Properties	Description
Input Fields	Displays the input fields that you had configured along with their associated values.
Fault	If the subtaskflow step faulted, you see the following properties: - Code. Indicates whether the fault is an error or a warning. - Reason. Indicates the cause of the error or fault. - Source. Provides details about the error or fault. If a Data Task step that uses a mapping task fails, the subtaskflow name followed by the mapping task name is displayed.
Output Fields	Displays the output fields of the subtaskflow along with their associated values.

Throw step properties

The Throw step properties display the fault details for the taskflow or the step that is associated with the Throw step.

If the step preceding the Throw step or the step with which the Throw step is associated faulted, the **Fault** field in the **Properties** panel displays the following details:

- **Code**. Displays the fault code.
- **Reason**. Displays the cause of the fault.
- **Details**. Displays the details about the fault.

End step properties

The End step properties display details about the current run of the taskflow.

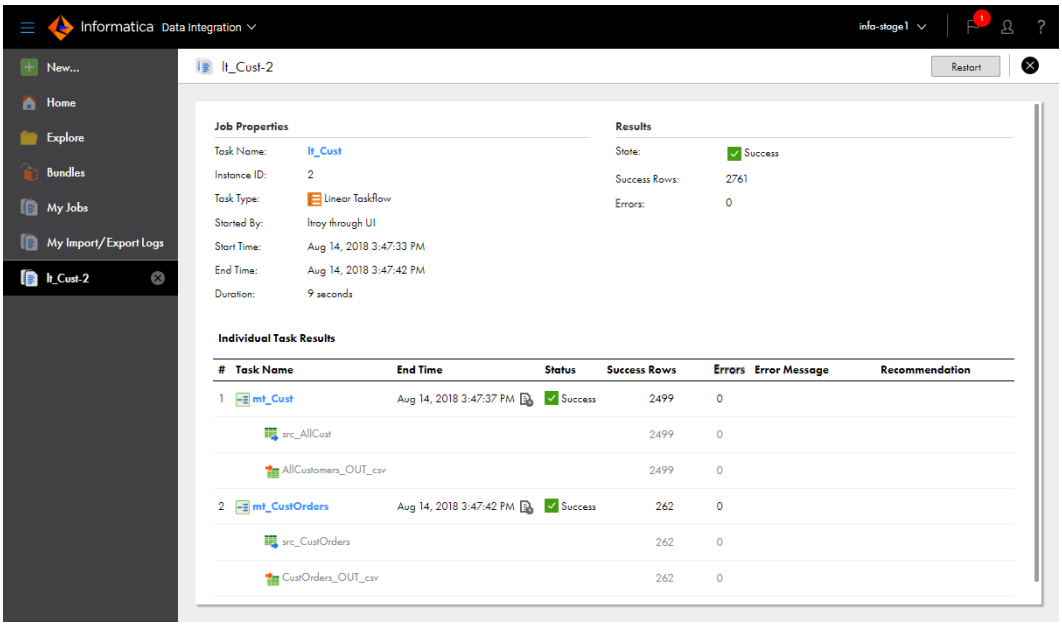
When you click an End step, the **Properties** panel displays the following information:

Properties	Description
Fault	If the End step faulted, you see the following properties: - Code. Displays the fault code. The code is set to **faultDueToNotRunOrFaulted** if the subtaskflow associated with the Subtaskflow step failed or did not run. - Reason. Indicates the cause of the error or fault. - Source. Displays the path of the step that faulted

Monitoring linear taskflows

To view detailed information about a linear taskflow instance, click the job name on the **My Jobs** page.

The following image shows the job details for a linear taskflow:



You view different details in each panel.

Job properties

The job properties for each linear taskflow instance display general properties about the instance.

The job properties include the following properties:

Property	Description
Task name	Name of the linear taskflow.
Instance ID	Instance number for the linear taskflow. For example, if you are looking at the third run of the taskflow, this field displays "3."
Task type	Task type, that is, Linear Taskflow.
Started by	Name of the user or schedule that started the job.
Start time	Date and time that the job was started.
End time	Date and time that the job completed or stopped.
Duration	Amount of time the job ran before it completed or was stopped.

Job results

The job results for each linear taskflow instance display the status of the job and success and error statistics.

The job results include the following properties:

Property	Description
Status	Job status. A job can have one of the following statuses: - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The job was stopped by a user or the parent job has stopped running, so the subtask can't start. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors.
Success rows	Number of rows successfully written to the target.
Errors	Number of rows that were not written to the target.

Individual task results

The individual task results display results for individual subtasks in the linear taskflow instance.

For each subtask, the job details include the following properties:

Property	Description
Task name	Name of the subtask.
End time	Date and time that the subtask completed or stopped.
Status	Status of the subtask. A subtask can have one of the following statuses: - Queued. The job is queued on a Secure Agent, but it has not started yet. - Starting. The job is starting. - Running. The job is still running. - Success. The job completed successfully. - Stopped. The subtask was stopped by a user or the parent job has stopped running, so the subtask cannot start. If you stop a subtask, Data Integration does not propagate the status to the parent task. - Warning. The job completed with errors. - Failed. The job did not complete because it encountered errors. - Aborted. The job was aborted. Applies to file ingestion and replication subtasks.
Success rows	Number of rows successfully written to the target. For each source and target in the subtask, this field displays the number of rows successfully read from the source and the number of rows successfully written to the target.
Errors	Total number of source error rows, target error rows, and transformation errors. For each source and target in the subtask, this field displays the number of rows that were not read from the source and the number of rows that were not written to the target.

Property	Description
Error message	Error encountered when writing to the target, if any.
Recommendation	Recommended action for errors.

Running only a particular step in a taskflow instance

You can run only a particular step in a taskflow instance from the **My Jobs** page.

When you run a particular step in a taskflow instance, Data Integration locks the entire taskflow so that other users can't run the taskflow. Data Integration runs only the selected step.

You can run only a particular step if the taskflow meets the following criteria:

- The taskflow step was run at least once.
- The taskflow is in one of the following states:
  - Success
  - Failed

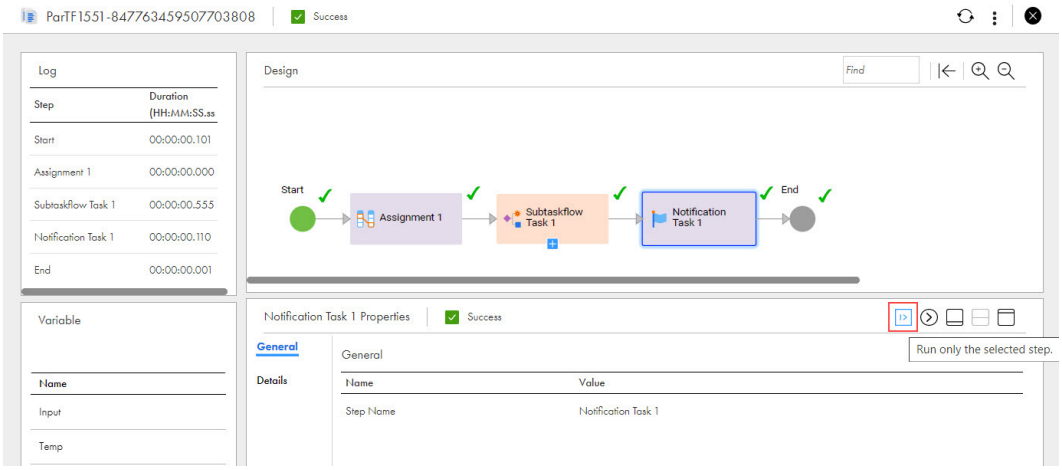
You can run a taskflow instance from any step other than the Start step, End step, and Jump step.

You can run only the following steps in a taskflow instance as a standalone step:

- Data Task step
- Ingestion Task step
- Command Task step
- IntegrationOps step
- Notification Task step
- File Watch Task step
- Subtaskflow step

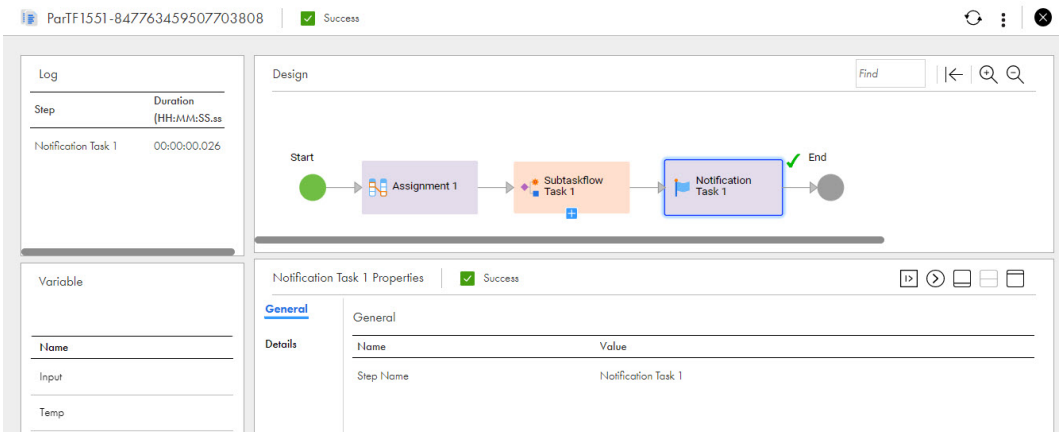
To run a particular step in a taskflow instance, open the **My Jobs** page, and then click the taskflow instance. On the job details page, select the step that you want to run and click the **Run only the selected step** icon.

The following image shows the **Run only the selected step** icon to run the selected Notification Task step in the taskflow instance:



Note: To run a selected step, specify both a source and a target location. If the Data Task uses custom error handling but you leave the error-handling path empty, only the source location passes, and the step fails with an error.

The following image shows the taskflow instance after it runs the selected Notification Task step:



When you run a failed task, if the task fails again at rerun, the overall status of the taskflow instance appears as failed on the job details page and on the **My Jobs** page .

Note: The taskflow ignores the error handling setting for the failed task if set to ignore on error.

Running a taskflow instance from a step

From the **My Jobs** page, you can run a taskflow from a particular step if the taskflow meets the following criteria:

- The taskflow step was run at least once.
- The taskflow is in one of the following states:
  - Success
  - Failed

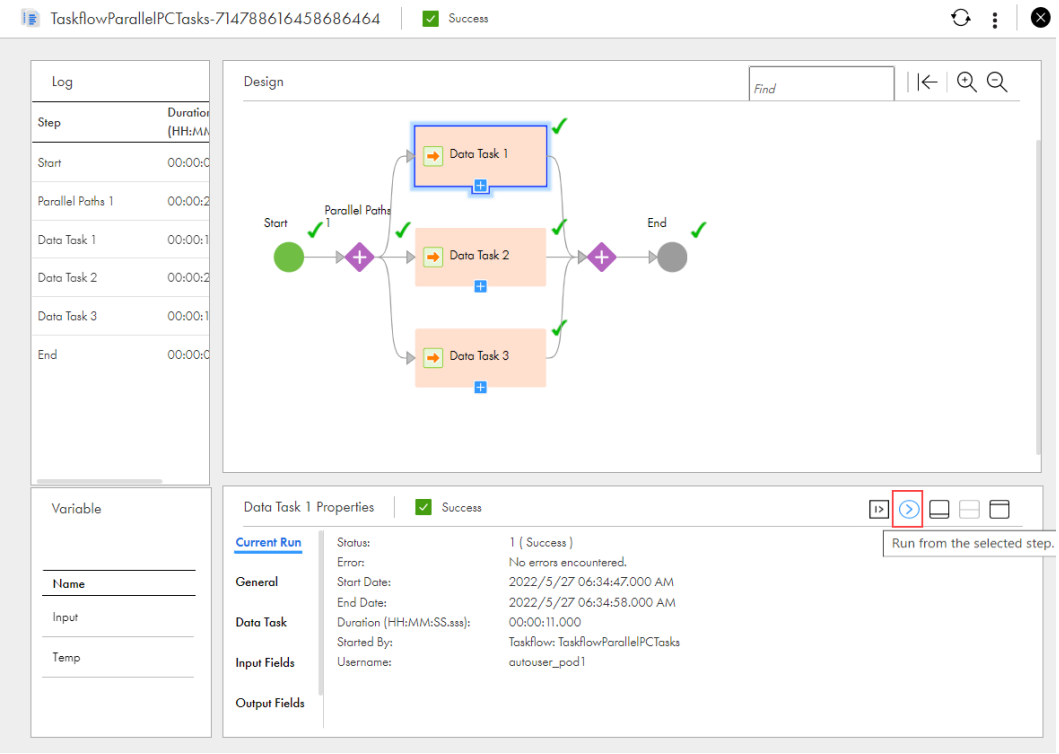
You can run a taskflow instance from any step other than the Start step, End step, Throw step, and Jump step.

When you run a taskflow instance from a particular step, if any subsequent step uses input from a previously executed step, it will have null values. To avoid this issue, update the input and temporary variables before you run a taskflow instance from a step. For more information about updating variables, see "Updating variables to run a taskflow instance from a step" on page 54.

To run a taskflow instance from a step, open the **My Jobs** page, and then click the taskflow instance. On the job details page, select the step from which you want to run the taskflow and click the **Run from the selected step** icon.

Note: If you update a taskflow, you must save and run the taskflow at least once before you can run the taskflow from a particular step.

The following image shows the **Run from the selected step** icon to run the taskflow instance from the selected Data Task step:



Note: When you run a taskflow from a particular step, the taskflow status might change. If the taskflow is used in another taskflow, the parent taskflow will not be impacted by the updated taskflow status. You can run a taskflow from the steps that ran successfully. You can also run a taskflow from a failed task such as a Data Task step, Ingestion Task step, or Command Task step provided the error handling for that task is set to ignore on error.

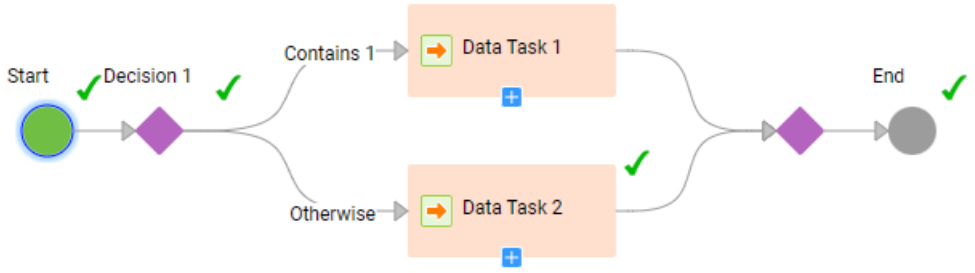
Updating variables to run a taskflow instance from a step

Use the **Variables** tab to update the input and temporary variables or fields and their associated values to run a complex taskflow instance from a particular step.

The **Variables** tab contains all the input variables and fields that are configured in the Start step for the taskflow. It also contains the temporary variables and fields that also include tasks such as Data Task step, Ingestion Task step, and Command Task step that were used to run the taskflow.

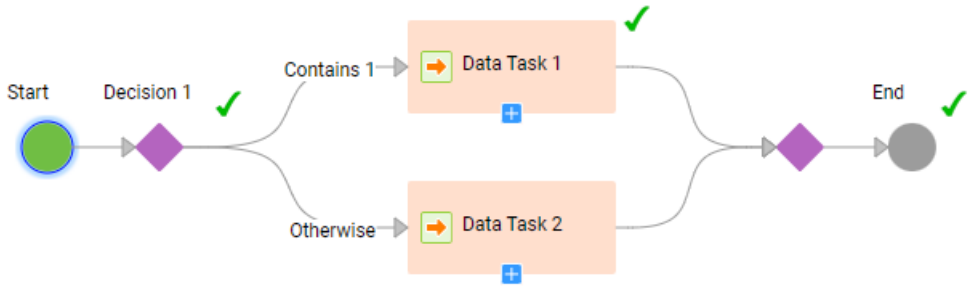
To update the input and temporary variables or fields and their associated values, open the **My Jobs** page, and then click the taskflow instance. On the job details page, click the **Edit** icon next to the Input or Temp variable. Based on the values passed in the taskflow, the variables are displayed in the JSON or XML format in the **Variable Input** and **Variable Temp** dialog boxes respectively. Update the variables and click **Update**.

You cannot run a task that never ran. For example, you cannot run a taskflow instance from the Data Task 1 step as shown in the following image:



In this case, you must update the variable value that is used in the Decision 1 step, and then run from the Decision 1 step again.

The following image shows the taskflow instance after it runs from the Decision 1 step:



For more information about the input and temporary fields, see the Taskflows documentation.

Stopping and restarting mapping and task instances

You can stop and restart mapping and task instances on the **My Jobs** page.

You can perform the following actions:

Stop a mapping or task instance.

You can stop a mapping or task instance that is starting, running, or queued.

Stop the job using one of the following methods:

- Clean Stop. Gracefully shuts down the DTM process.
- Stop. Forcefully stops the DTM process.

To stop a mapping or task instance, open the **My Jobs** page. Then, click the **Clean Stop** icon or the **Stop** icon in the row that contains the job that you want to stop.

When you click **Clean Stop**, the job status changes to Stopping. If the job takes too long to stop, click **Stop** to stop the job.

Restart a mapping or task instance.

You can restart a mapping or task instance that has run successfully or failed.

To restart a mapping or task instance on the **My Jobs** page, click the **Restart** icon in the row that contains the job that you want to restart.

To restart a mapping or task instance from the job details, click **Restart** at the top of the page.

Stopping, suspending, resuming, and restarting taskflow instances

You can stop, suspend, resume, and restart taskflow instances on the **My Jobs** page.

You can perform the following actions:

Stop a taskflow instance.

You can stop a taskflow instance that is in the suspended or running state.

To stop a taskflow instance, open the **My Jobs** page, and then click the **Stop** icon in the row that contains the taskflow instance.

You can also stop the taskflow instance from the job details page. To stop a taskflow instance, open the **My Jobs** page, and then click the taskflow instance. On the job details page, click the **Stop** icon at the top of the page.

When you stop a parent taskflow, all the underlying subtaskflows that are running also stop. The **Last Fault Name** field in the **Taskflow Properties** > **Current Run** section of the parent taskflow that you stop displays the value as **processManuallyTerminated**. However, the **Last Fault Name** field of the subtaskflows displays the value as **processTerminated**.

Suspend a taskflow instance.

To suspend a taskflow instance, open the **My Jobs** page, and then click the **Suspend** icon in the row that contains the taskflow instance.

Resume a taskflow instance from a faulted step.

To resume a suspended taskflow instance from a faulted step, open the **My Jobs** page, and then click the taskflow instance. On the job details page, click the **Resume from faulted step** icon. The taskflow resumes from the step that faulted.

If you resume a taskflow that was suspended because of a data integration task fault, the task re-runs. For information about suspending a taskflow on a data task fault, see *Taskflows* in the Data Integration service help.

Skip a faulted step and resume a taskflow instance from the next step.

To skip a faulted step and resume a suspended taskflow instance from the next step, open the **My Jobs** page, and then click the taskflow instance. On the job details page, click the **Skip faulted step and resume** icon. The taskflow skips the faulted step and resumes from the next step. If all the other steps run successfully, the taskflow status is set to success.

Restart a taskflow instance.

Users who have the Admin, Designer, or Operator role can restart a taskflow from Monitor and Data Integration.

You can restart a taskflow that meets the following criteria:

- The taskflow was run at least once.
- The taskflow is in one of the following states:
  - Success
  - Failed
  - Suspended

You can also restart a taskflow instance that has the **Binding** property set to **Event**.

To restart a taskflow instance, open the **My Jobs** page, and then click the **Restart** icon in the row that contains the taskflow instance.

CHAPTER 3

Data Integration job log files

Data Integration generates log files to help you monitor running, failed, and completed jobs. You can access some of the log files from the **My Jobs** pages, and from the job details.

Data Integration generates the following types of log files:

Error rows file

Data Integration generates error rows files for synchronization task and masking task instances. An error rows file shows the rows that failed and the reason why each row failed. The error rows file includes the first 50 fields of a source error row.

For example, the following error appears in the error rows file when the task tries to insert two records with the same external ID into a Salesforce target:

```
Error loading into target [HouseholdProduct__c] : Error received from salesforce.com.
Fields [ExternalId__c]. Status code [DUPLICATE_VALUE]. Message [Duplicate external id
specified: 1.0].
```

Session log file

Data Integration generates a session log file for each job. This log gives you a high level view of the time spent for different operations.

The session log provides mapping compilation time, translation time, simplification time, optimization time, total time to create the LDTM, Spark task submission time, Spark task [InfaSpark0] execution start and end time, and Total time to perform the LDTM operation.

If a job fails, analyze the session log file first to help you troubleshoot the job.

Reject file

Data Integration creates a reject file for each flat file and Oracle target in a mapping or mapping task that contains error rows. The reject file contains information about each rejected target row and the reason that the row was rejected. Data Integration saves the reject file to the following default folder:

```
$PMBadFileDir/<task federated ID>
```

Execution plan

Data Integration generates an execution plan that shows the Scala code that an advanced cluster uses to run the data logic in a mapping in advanced mode. You can use the Scala code to debug issues in the mapping.

Agent job log

Data Integration generates an agent job log that shows the logic that the Secure Agent uses to push the Spark execution workflow for a mapping in advanced mode to an advanced cluster for processing.

The agent job log contains information such as metering, time the application was submitted to the cluster, and the time the application completed. This log can help you troubleshoot delays in running the Spark task in the Session log, and you can see when the Spark task was processed on the Secure Agent.

Spark driver and Spark executor logs

An advanced cluster generates Spark driver and Spark executor logs to show the logic that the cluster uses to run a job. Use these logs to identify issues or errors with the Spark process. This log also contains information about the different executors being created and the tasks that are starting or have been completed.

Initialization script log

If an initialization script runs on an advanced cluster, the cluster generates an init script log to show the script output.

Cloud-init log

If an initialization script runs on the advanced cluster, the cluster generates a cloud-init log that contains information about how cluster nodes were initialized and bootstrapped. You can use the cloud-init log to check if any init scripts failed to run.

Note: You can view the cloud-init log only in an AWS environment.

Spark event log

An advanced cluster generates a Spark event log to stream runtime events for tasks that run on the cluster.

The Spark event log records different events in a JSON-encoded format while the application is running. This log contains the events associated with the Spark application, such as the different jobs in the application, different stages, individual tasks, and interaction between entities.

The Spark event log also contains events related to the software infrastructure like driver information, executor creation, memory usage by executors, environment configuration, and the logical and physical plans of the Spark application. Use this log to trace what happened during every step of the Spark application run.

To find the Spark event log, open the Spark driver log and search for `SingleEventLogFileWriter`. The result of the search shows the path of the Spark event log. For example:

```
23/01/09 04:38:35 INFO SingleEventLogFileWriter - Logging events to s3://bucket/
log_location_in_cluster_condifuration/eventLogs/atscaleagent/spark-
a7bea557ede14382b4807d35b5404b97.inprogress
```

When the application completes, download the Spark event log from the location `s3://bucket/log_location_in_cluster_condifuration/eventLogs/atscaleagent/` as file `spark-a7bea557ede14382b4807d35b5404b97`.

To interpret the Spark event log, import it into a Spark history server and examine the log using the History server monitor. Check the following tabs:

- The **Jobs** tab shows all the detailed metrics.
- The **Stages** tab lists all the completed stages. You can see detailed information on the total number of tasks succeeded or failed, input and output data volume, and shuffle read and shuffle write data volume. Click on any stage to view the DAG visualization diagram.
- The **Environments** tab shows the Spark-related parameters used to run the Spark job.
- The **Executors** tab shows detailed information about the executor Pods and driver Pod.

For more information, refer to the Apache Spark documentation.

Advanced logs

The Advanced Log Location contains the Spark executor logs apart from the Spark driver and Agent job logs. The executor logs can help you troubleshoot issues with individual executors.

Viewing an error rows file

You can view an error rows file using one of the following methods:

Preview error rows file

Preview the error rows file from the **My Jobs** page or from the job details. When you preview the error rows file, Data Integration generates a CSV file that contains the first 25 error rows and first 50 columns.

To preview an error rows file, your user role must have the "Access CDI Error Logs" feature privilege for Data Integration.

Select **My Jobs**.

Perform either of the following steps:

- Click the **Preview Error Rows File** icon in the row that contains the task instance.
- Click the job name to open the job details, and then click **Preview Error Rows File** in the Results area.

View error rows file

To view the entire error rows file for a job, open the file in a text editor. If the error rows file contains Unicode data from the source, open the error rows file in an application that can display Unicode characters.

Error rows files are written to the following directory:

```
<Secure Agent installation directory>/apps/Data_Integration_Server/data/error
```

Error rows files are named with a timestamp that identifies the start time of the job. To identify the correct error rows file, use the start time of the job.

Downloading a session log file

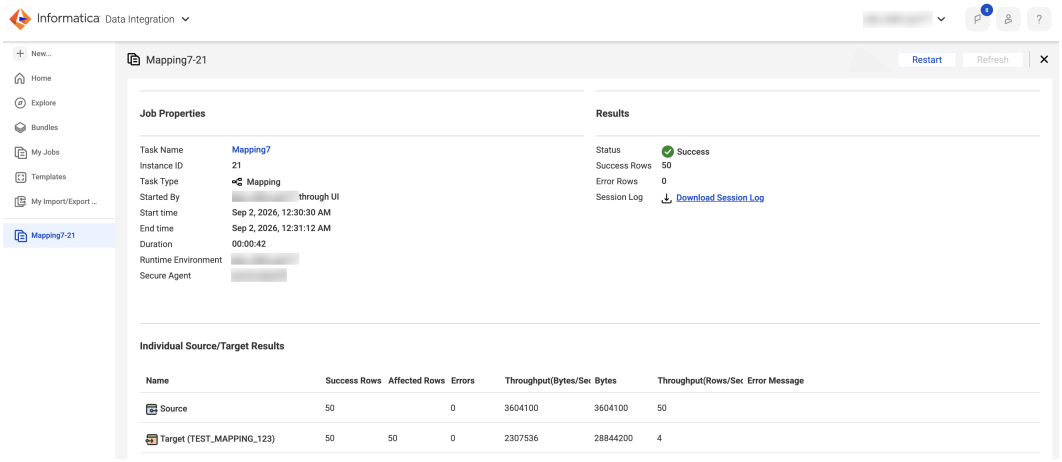
You can download the log file for a job that is running, completed, or failed. Download the log file from the **My Jobs** page or from the job details.

1. Select **My Jobs**.
2. Perform either of the following steps:
   - Click the **Download Log** icon in the row that contains the job.
   - Click the job name to open the job details, and then click **Download Session Log** in the Results area.

Auditing Access Policy transformations

When you run a mapping with an Access Policy transformation, you can audit the rules that the transformation applies to your data and audit the columns or rows that the rules apply to. Find the audit information in the session log, You can download the session log from the My Jobs page.

The following image shows the job properties of a mapping with a link to download the session log:



After you select an instance of the mapping execution, you can download the session log. The session log displays the rules applied and the columns or rows that the rules were applied to.

The following image shows a session log:



The log includes information about the following types of rule conditions that the transformation applies to the data:

1. Field-level conditions of a data de-identification rule
2. Cell-level conditions of a data de-identification rule
3. Row-level conditions of a data filter rule

Viewing log files for advanced cluster subtasks

When you monitor a subtask that runs on an advanced cluster, you can view the execution plan, the session log, the agent job log, and the Spark driver and Spark executor logs. If an initialization script runs on the cluster, you can also view the init script logs and cloud-init logs.

To access a specific log file for the job, you might have to download the log from Data Integration or from your cloud platform.

The following table indicates whether you can download each log from Data Integration, your cloud platform, or both:

Log	Download location
Execution plan	Data Integration
Session log	Data Integration
Agent job log	Data Integration, cloud platform[1]
Spark driver log	Data Integration, cloud platform[1]
Spark executor log	Data Integration[2], cloud platform[1]
Initialization script log	Cloud platform
Cloud-init log	Cloud platform
Spark event log	Cloud platform

1 *If you use a serverless runtime environment, downloading from your cloud platform is not available.*

2 *If you do not use a serverless runtime environment, downloading from Data Integration is not available. Use the displayed log location to download the log from your cloud platform.*

Downloading the execution plan and session log

You can download the execution plan and the session log from the job results in Data Integration.

Downloading advanced logs

You can download the agent job log and the Spark driver log in the Spark task details in Data Integration. You can also copy the advanced log location and navigate to the location on your cloud platform. You must navigate to the advanced log location to download Spark executor logs.

If you use a serverless runtime environment, you can download the advanced logs from Data Integration. The **Download** option becomes available when the job completes.

Downloading init script and cloud-init logs

To download an initialization script log or a cloud-init log, navigate to the following location on your cloud platform:

```
<Log location>/ClusterLogs/<cluster ID>/<timestamp>/<cluster node instance ID>/
```

The log location is the location for log files that is specified in the advanced configuration.

Note: You can download cloud-init logs only in an AWS environment.

Downloading Spark event logs

To download the Spark event log, navigate to the following location:

```
<Log location>/eventLogs/atscaleagent
```

The log location is the location for log files that is specified in the advanced configuration.

If the job fails to start, you see the following sample text in the event log file:

```
{"Event":"SparkListenerLogStart","Spark Version":"3.0.0"}
```

For more information about the job failure, see the Spark driver log.

Rules and guidelines

Refer to the following rules and guidelines when you view log files:

- If you download the Spark driver and agent job logs while the job is running, the files reflect the current state of the job.
- If you view a log file for a subtask that runs on an advanced cluster with a mapping name that begins with a number or a space, the mapping name is prefixed with an underscore in the log file.

  For example, when a log file references a mapping named `123Mapping`, the log files use the name `_123Mapping`.

Taskflow log files

You can download the log file for a taskflow to review it offline and troubleshoot problems, if necessary. The downloaded log file name contains the name of the taskflow, the run ID, and the organisation ID of the user who triggered the taskflow.

Note: If the taskflow name contains a space, the log file name contains a **+** symbol.

Downloading a taskflow log file

You can download a taskflow log file to review it offline and troubleshoot problems.

1. Open the **My Jobs** page.
2. Click the taskflow in the **Instance Name** column to open the corresponding taskflow details page.
3. In the **Results** area, click the **Download Log** icon to download the taskflow log file.

Downloading a taskflow log file using the log resource

After you publish a taskflow as a service, you can run the taskflow and download the log file for the taskflow log file using the log resource.

You can use one of the following ways to invoke the request through a REST client such as Postman:

- Use basic authentication, and enter the user name and password.
- Use the session ID in the HTTP header to invoke the taskflow without providing the user name and password.

For example, you can authenticate the GET request in one of the following ways:

- Use basic authorization and specify the IDMC user name and password.

For example:

```
GET<IDMC URL>/active-bpel/services/tf/log/<run ID>
Accept: application/octet-stream
Authorization : Basic Auth
username: <IDMC username>
password: <IDMC password>
```

- Use the IDS-SESSION-ID in the HTTP header.
  For example:

  ```
  GET<IDMC URL>/active-bpel/services/tf/log/<run ID>
  Accept: application/octet-stream
  IDS-SESSION-ID : <sessionId>
  ```

  To get the SESSION-ID, use the Platform REST API version 3 login resource. For more information about the login resource, see *REST API Reference*.

GET request

To download a log file for the taskflow, use the following URI:

```
GET<IDMC URL>/active-bpel/services/tf/log/<run ID>
```

For example:

```
https://na4.dm.us.informaticacloud.com/active-bpel/services/tf/log/20262247166322413568
```

Taskflow log file contents

The header of the log file contains information such as the time stamp when the file was downloaded, user ID, and the status of the taskflow at the time of downloading the log file. The body of the log file contains the following information:

Field	Type	Description
assetName	String	Name of the taskflow. The taskflow name also includes the custom name, if you had added a custom name to the taskflow using an API or the RunAJob utility.
assetType	String	Type of the object. Returns the value TASKFLOW.
duration	String	Time in seconds that the taskflow ran before it completed, was suspended, was failed, or was stopped.
endTime	Date/time	End time for the taskflow run. Uses Coordinated Universal Time (UTC).
location	String	Project and folder path where the taskflow is saved.
runId	Long	Run ID for the taskflow.
URL	String	The log resource URL generated by Data Integration. The log file will contain one URL corresponding to the taskflow and as many URLs as the subtasks within the taskflow.
runtimeEnv	String	The name of the application node where the taskflow runs.
runtimeEnvName	String	Name of the runtime environment where the taskflow runs.
startTime	Date/time	Start time for the taskflow run. Uses Coordinated Universal Time (UTC).
startedBy	String	Name of the user who first started the taskflow.

Field	Type	Description
status	String	Execution status of the taskflow. Returns one of the following values: - RUNNING. The taskflow is running. - SUCCESS. The taskflow completed successfully. - FAILED. The taskflow did not complete because it encountered errors. - SUSPENDED. The taskflow run was suspended.
subtasks	String	Number of subtasks that the taskflow contains.
updateTime	Date/time	Last time the taskflow run status was updated. Uses Coordinated Universal Time (UTC).
statusCode	String	Status code of the taskflow. The status code can be one of the following values: - 1 - Running or CHILD_SUSPENDED - 2 - Suspended - 3 - Success - 4 - Failed - -1 - Default
errorMessage	String	Error message string.
subtaskDetails	String	Object that contains status details for all subtasks in the taskflow.
details	String	Status details. Includes status information for each subtask in the tasks object.
tasks	Collection	Status information for all subtasks that the taskflow contains.

Sample taskflow log file

The content of the log file is of octet-stream type. The following snippet shows a sample of the downloaded log file for a taskflow:

```
{
    [Tue Aug 29 09:49:11 UTC 2023] [sadityaRelPod1] -  TASKFLOW
TaskflowRelease15467983  FAILED.
Details:
{
 "assetName" : "TaskflowRelease15467983",
 "assetType" : "TASKFLOW",
 "duration" : 795,
 "endTime" : "2023-08-29T06:05:44Z",
 "errorMessage" : "{\"reason\":\"Unknown\",\"code\":\"Unknown\",\"details\":\"Unknown
\"}",
 "location" : "Default",  "runId" : 881053605371789312,
 "URL" : "https://qa-pod1.rel.infaqa.com/active-bpel/services/tf/log/881053605371789312",
 "runtimeEnv" : "taskflow-qa-release-pod1-r41-app02.infacloudops.net:4430",
 "runtimeEnvName" : "",
 "startedBy" : "sadityaRelPod1",
 "startTime" : "2023-08-29T05:52:29Z",
 "status" : "FAILED",
 "subtasks" : 1,
 "updateTime" : "2023-08-29T06:05:44Z",
 "statusCode" : 4,
 "subtaskDetails" : {
 "details" : {
 "tasks" : [ {
 "assetName" : "MappingTask2",
 "assetType" : "MTT",
 "duration" : 13,
 "endTime" : "2023-08-29T05:52:43Z",
```

```
 "errorMessage" : "",
 "errorRows" : 0,
 "location" : "Shivendra",
 "rowsProcessed" : 6,
 "runId" : 5,
 "runtimeEnv" : "01589925000000000002",
 "runtimeEnvName" : "INWPG02P31F-AAD",
 "startedBy" : "sadityarelpod1",
 "startTime" : "2023-08-29T05:52:30Z",
 "status" : "COMPLETED",
 "subtasks" : 0,
 "successRows" : 6,
 "updateTime" : "2023-08-29T05:52:44Z"      } ]    }  }}
    }
}
```

CHAPTER 4

Monitoring imports and exports

You can monitor the imports and exports that are running or have run in your organization. Each time that you import or export objects, IDMC creates a log entry for the import or export.

You can view the import and export logs on the following pages:

My Import/Export Logs page

Lists all imports or exports that were started by the currently logged in user. If you view your exports, you can download the export files from this page.

To open the **My Import/Export Logs** page, select **My Import/Export Logs**.

Details page

Displays detailed information about a specific import or export instance. You can also download the import or export instance log file. The log contains instance and object-level details. Logs are available for download for seven days. If you view details for an export instance, you can download the export file from this page.

To open the details page, click an import or export instance on the **My Import/Export Logs** page.

Import and export log properties

To view properties for an import or export instance, click the **Import** or **Export** tab at the top of the **My Import/Export Logs** page. By default, these pages list the import and export instances that were started within the last 7 days.

The following properties are displayed for each import or export instance:

Property	Description
Instance name	Name of the import or export instance.
Start time	Date and time that the import or export was started.
User name	Name of the user that started the import or export.

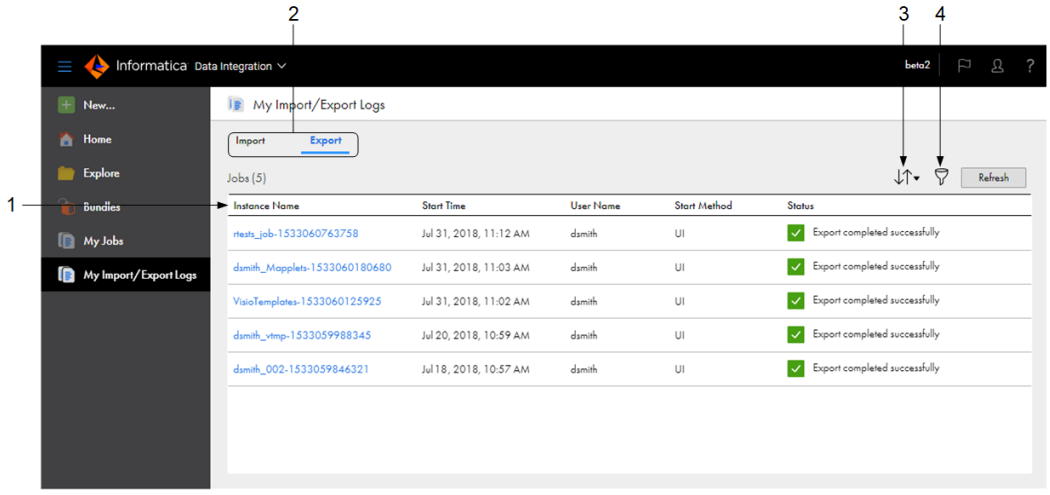
Property	Description
Start method	Method in which the import or export was started. An import or export instance can have one of the following start methods: - UI. The import or export was started from the user interface. - API. The import or export was started from the REST API.
Status	Status of the import or export, such as completed successfully or failed.

For successful export jobs, you can download the export file by clicking the **Download** icon in the row that contains the export instance.

Customizing import and export logs

You can decide which properties to display on the **My Import/Export Logs** page. You can sort and filter import or export instances. You can also drill down on an import or export instance to view detailed information.

The following image shows the properties that are displayed by default in Data Integration:



1. Column heading area. Right-click this area to add and remove columns.
2. Tabs to display either import logs or export logs
3. Sort icon
4. Filter icon

You can customize the import and export logs in the following ways:

Display, hide, or rearrange properties.

To display or hide specific properties, right-click the column heading area and check or uncheck the properties.

To rearrange the columns, click a column heading and drag it to a different location.

Sort import or export instances.

To sort the import or export instances, click the column heading for the property that you want to sort by. For example, to list the most recent jobs first, click the Start Time column. The arrow in the column heading indicates the sort order, either ascending or descending. To reverse the sort order, click the column heading again.

You can also sort import or export instances by clicking the **Sort** icon and selecting the column name.

Filter import or export instances.

You can filter import or export instances using one or more filter conditions. To filter the instances that appear on the page, click the **Filter** icon.

To specify a filter condition, click **Add Field**, select the filter field, and then enter the value. For example, to view only successful exports or imports, select the Status column, and select “Success.” You can select Instance Name or Status as the filter field.

To add another condition to the filter, click **Add Field** again. For example, to view failed instances named "SalesProject_1533326631921," select the following filter columns and values:

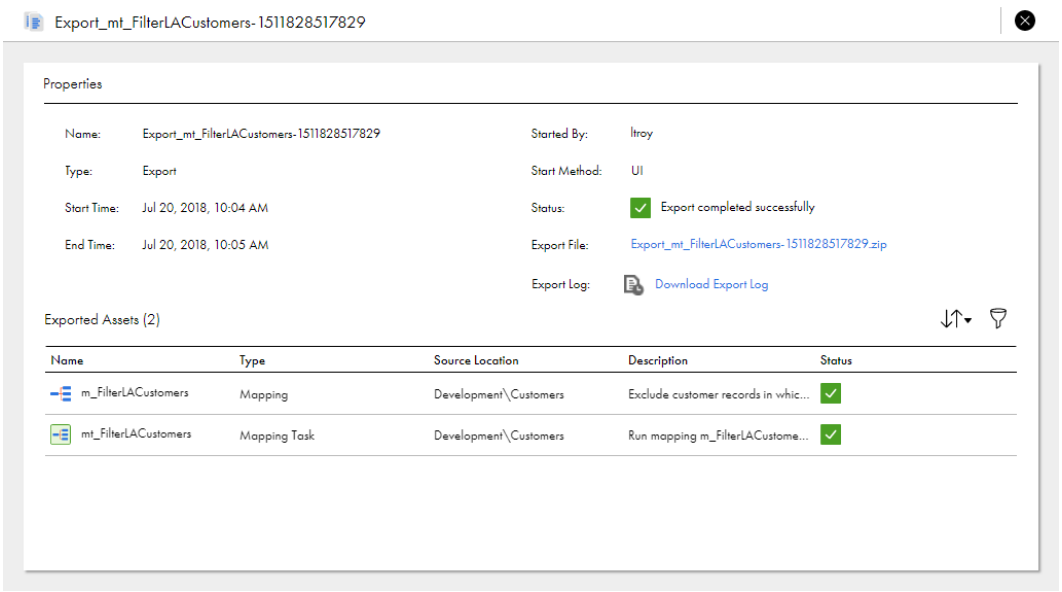
- Instance Name: SalesProject_1533326631921
- Status: Failed

To remove all applied filters, click the **Remove Filter** icon.

Viewing details for an import or export

To view detailed information about an import or export, click the **Import** or **Export** tab at the top of the **My Import/Export Logs** page. Then click the import or export instance name.

The following image shows the details for an export instance:



The details include the following properties:

Property	Description
Name	Name of the import or export instance.
Type	Instance type, either Import or Export.

Property	Description
Start time	Date and time that the import or export was started.
End time	Date and time that the import or export completed or stopped.
Started by	Name of the user that started the import or export.
Start method	How the import or export was started. An import or export can have one of the following start methods: - UI. The import or export was started from the user interface. - API. The import or export was started from the REST API.
Status	Status of the import or export, such as completed successfully or failed.
Source organization	For imports, the name of the organization from which assets were imported.
Export file	For exports, name of the export file that was generated. Click the file name to download the file.
Import log / Export log	Detailed log file containing instance and object-level properties. Click **Download Import Log** or **Download Export Log** to download the file.
Imported assets / Exported assets	Name, type, location, description, and status of each asset that is imported or exported. The imported assets or exported assets list is available for one day.

Downloading an export file

When an export has completed, you can download the export file. Download the export file from the **My Import/Export Logs** page or from the export instance details.

1. Select **My Import/Export Logs** to open your import/export logs.
2. Click the **Export** tab.
3. Perform either of the following steps:
   - Click the **Download** icon in the row that contains the export instance.
   - Click the instance name to open the export instance details, and then click the export file name.

Downloading an import or export log

When an import or export instance is complete, you can download the log file. Download the log file from the import or export instance details.

1. Select **My Import/Export Logs** to open your import/export logs.
2. Click the **Import** or **Export** tab.
3. Click the instance name to open the import or export instance details.

4. Click **Download Import Log** or **Download Export Log**.

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